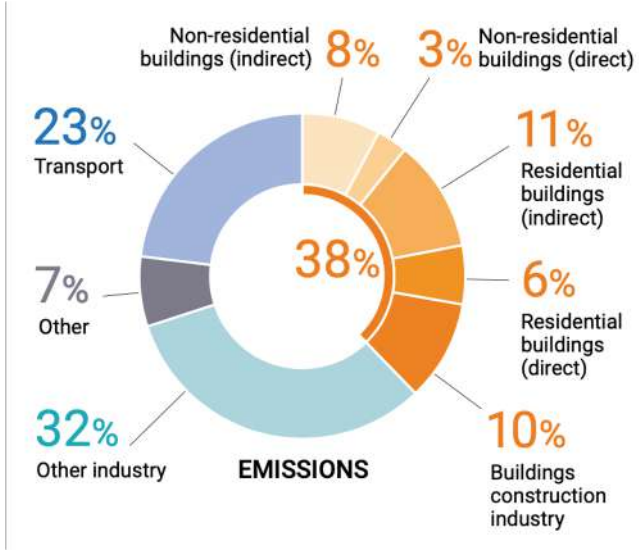

AMBIENT
SYSTEMS
Reimagining the future of smart cities

40%

of global energy and CO2 emissions is attributed to buildings

Emissions vs Productivity





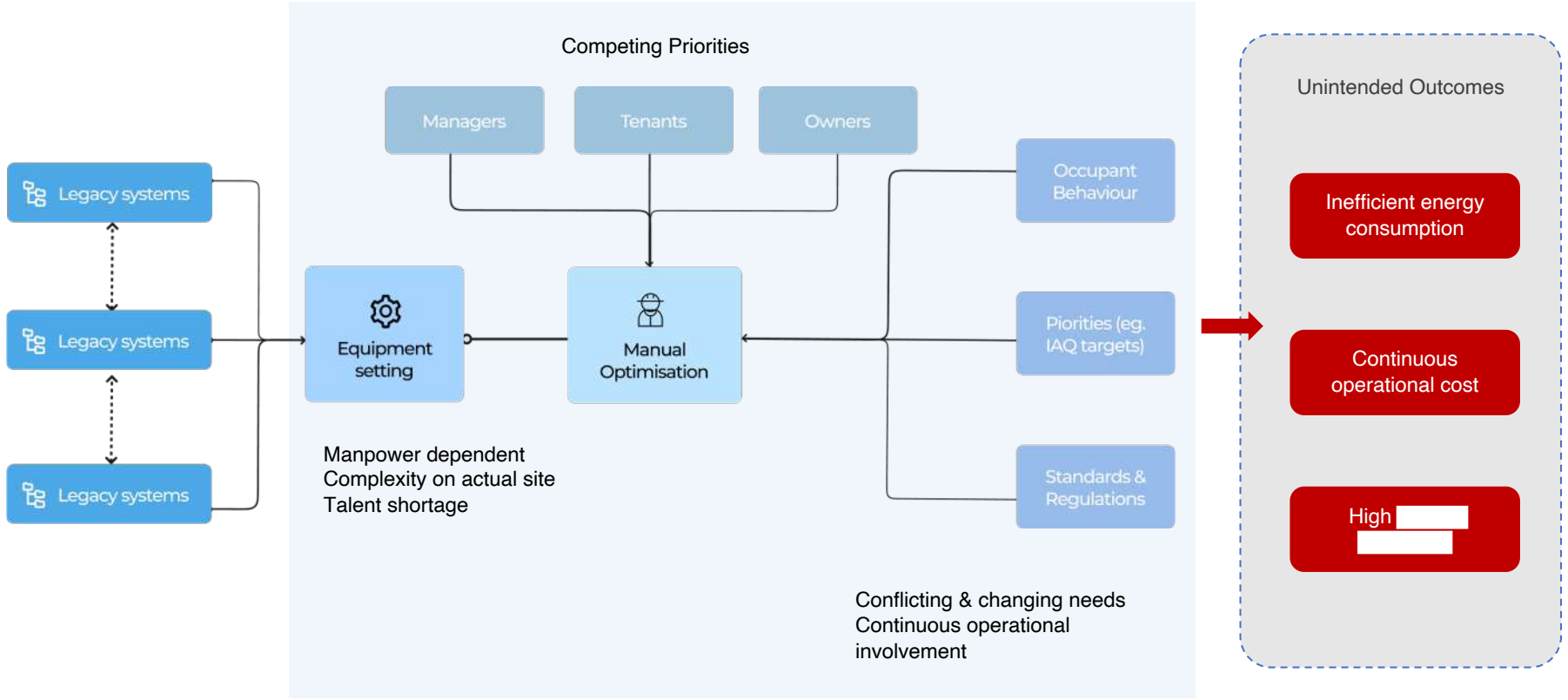



CapitaLand
Reimagine:
Decarbonize your CRE portfolio while driving innovation forward

Data Science

A missing link in CRE optimization & decarbonization

Decarbonization & Optimization in Buildings is Complex & Cumbersome



Hard to Find Expertise for Managing Net-Zero Smart Buildings

Edge Devices

Automation

User Behavior

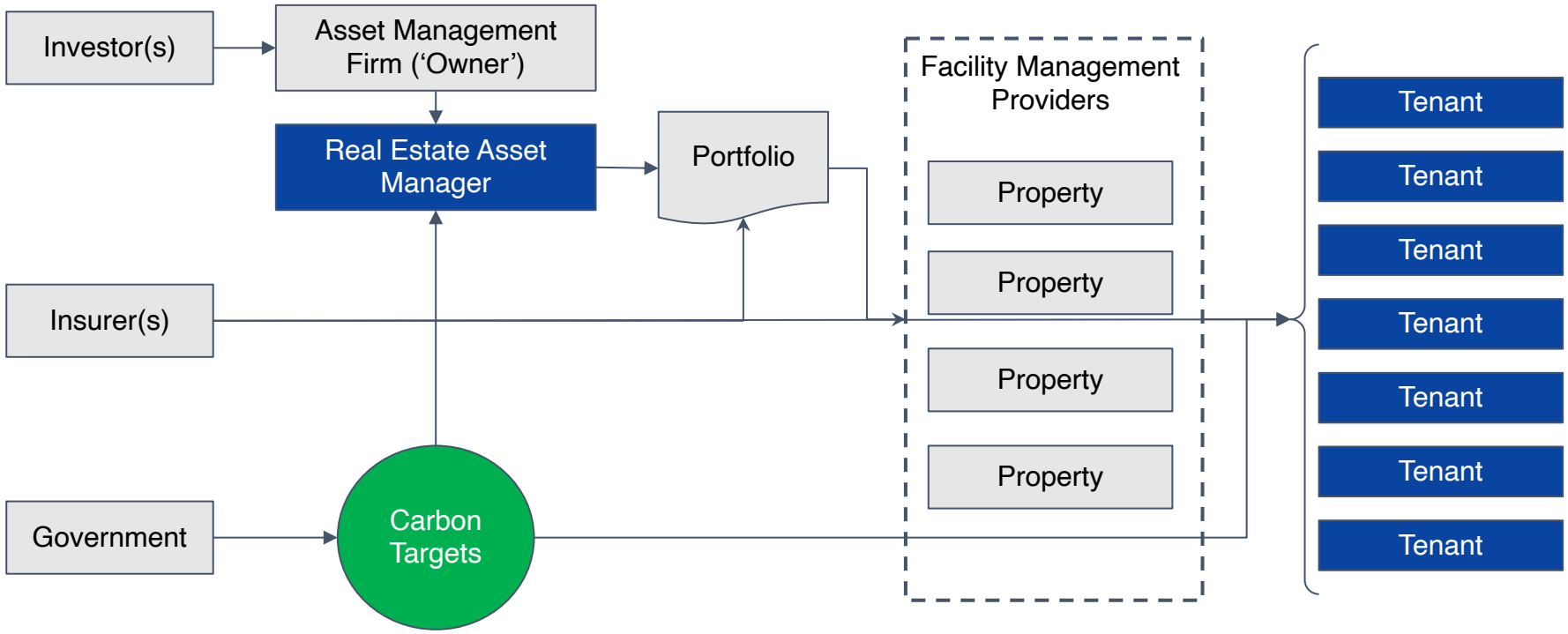


Inefficient energy consumption

Continuous operational cost

High

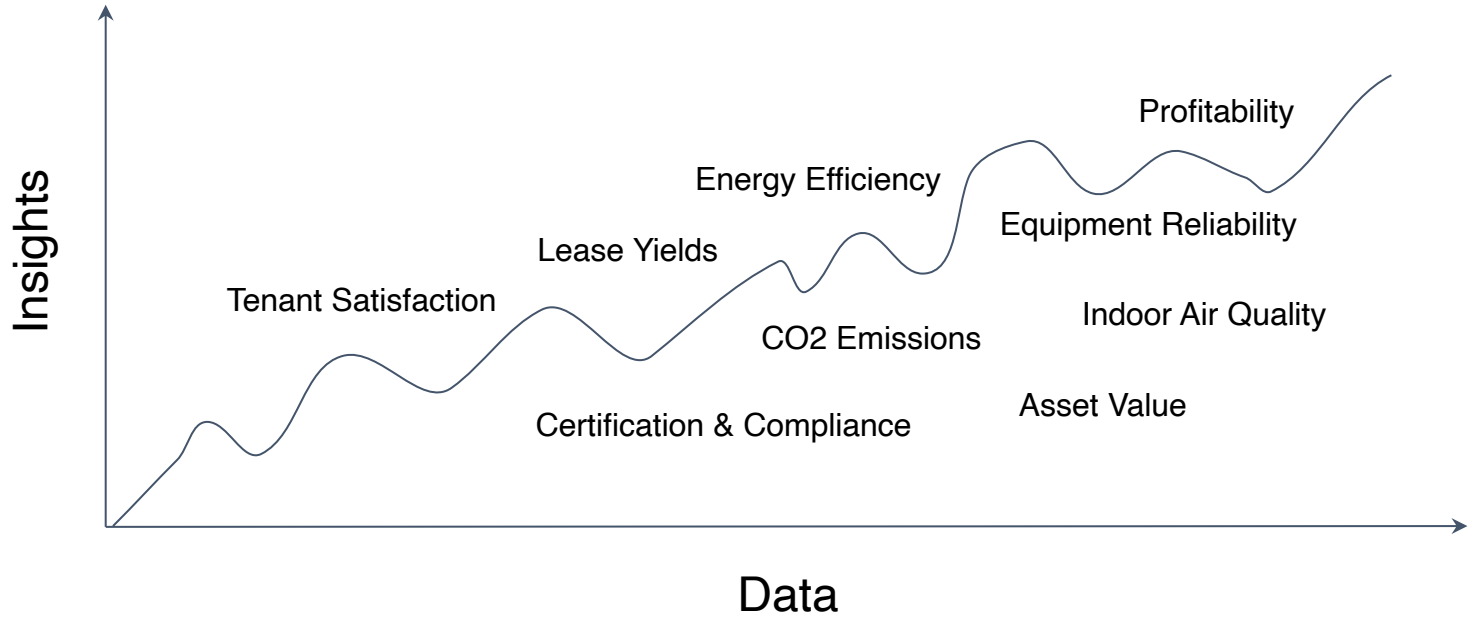
Complex Stakeholder Management Process



Gap Between Property Managers and Asset Managers

Property Management	Asset Management
Driven by operational and functional matters	Driven by financial matters
• Oversee and manage daily operations • Hire key facility management personnel • Orchestrating maintenance & repair tasks • Energy management • Indoor Air Quality • BMS / BAS operations	• Maxime property value and returns • Hire key personnel including property managers • Work with leasing agents • Communicating with investors and clients (occupiers) • Limit risks & liabilities

Data Science: empowering asset managers and their teams to connect the dots



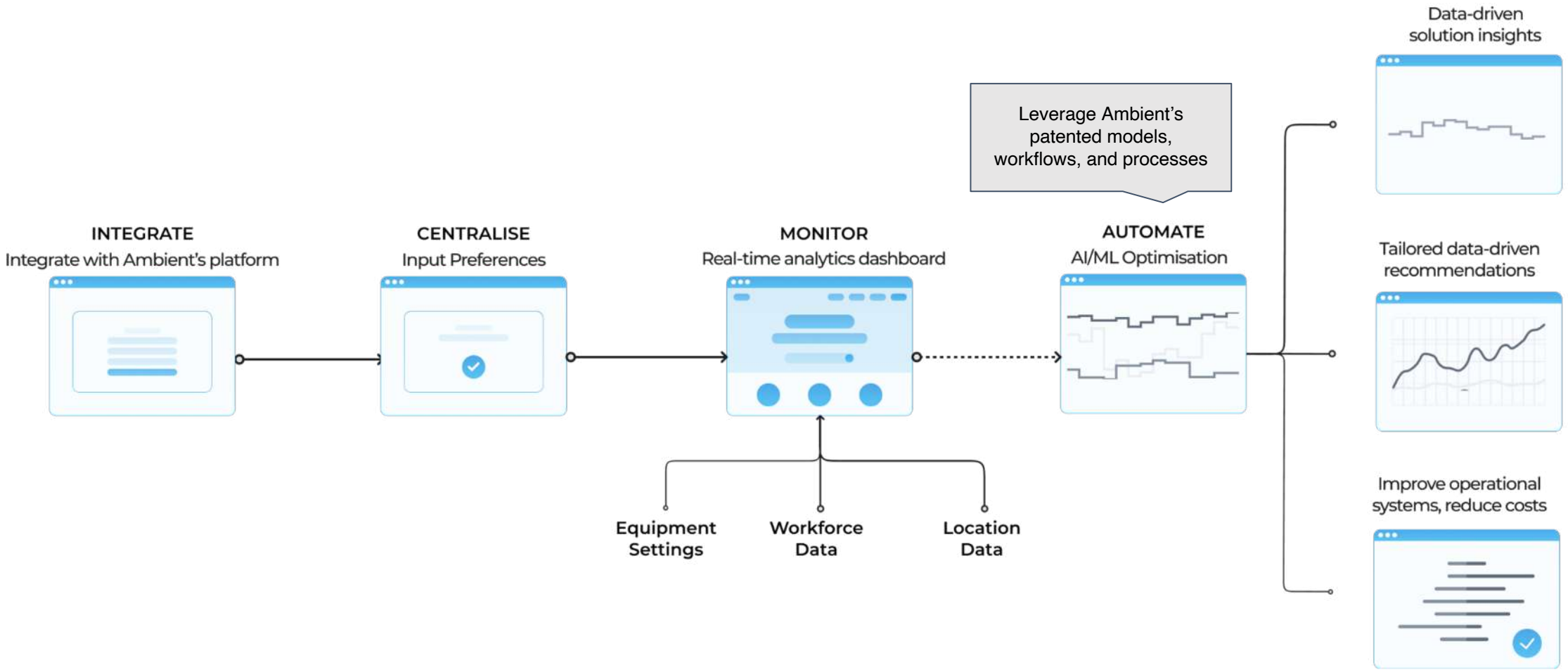
Ambient Systems

Product & Solution Offering

Platform for CRE Optimization & Decarbonization

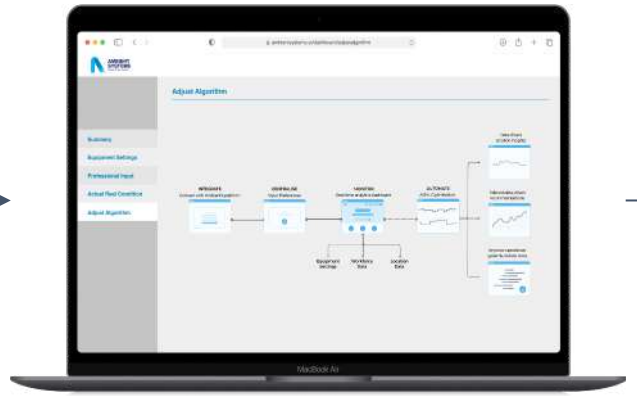
Streamlined management of property operations with Ambient's state of the art platform and proprietary data science methodology

Data Science pipeline for Real Asset Optimization & Decarbonization



Not a BMS or BAS but Compatible with your and Protocols

Ambient helps your facility managers make sense of BMS data and streamline



BMS Database

Siemens Building Technologies, , Honeywell, Bosch, ABB, Schneider Electric

Balanced CRE optimization & decarbonization through three unique features



Flexible On-demand Subscription

No more onboarding, training and manpower reliance. Fast up-time and results through Ambient's subscription service



Simple and Affordable Pricing Plans

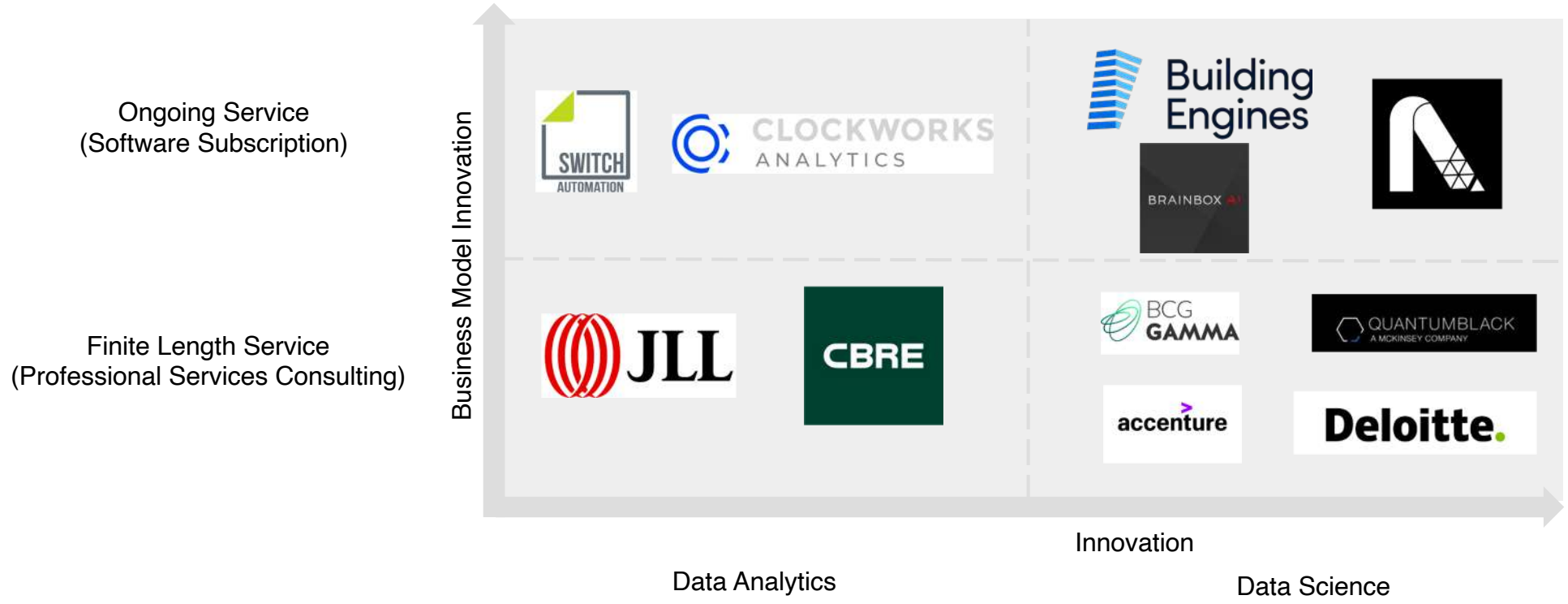
No more reliance on expensive consultants or in-house teams. One platform powered for all your data science use cases



Specialized AI Model Customization

Reach your business goals across energy efficiency, environment, and compliance with Ambient's state of the art AI models

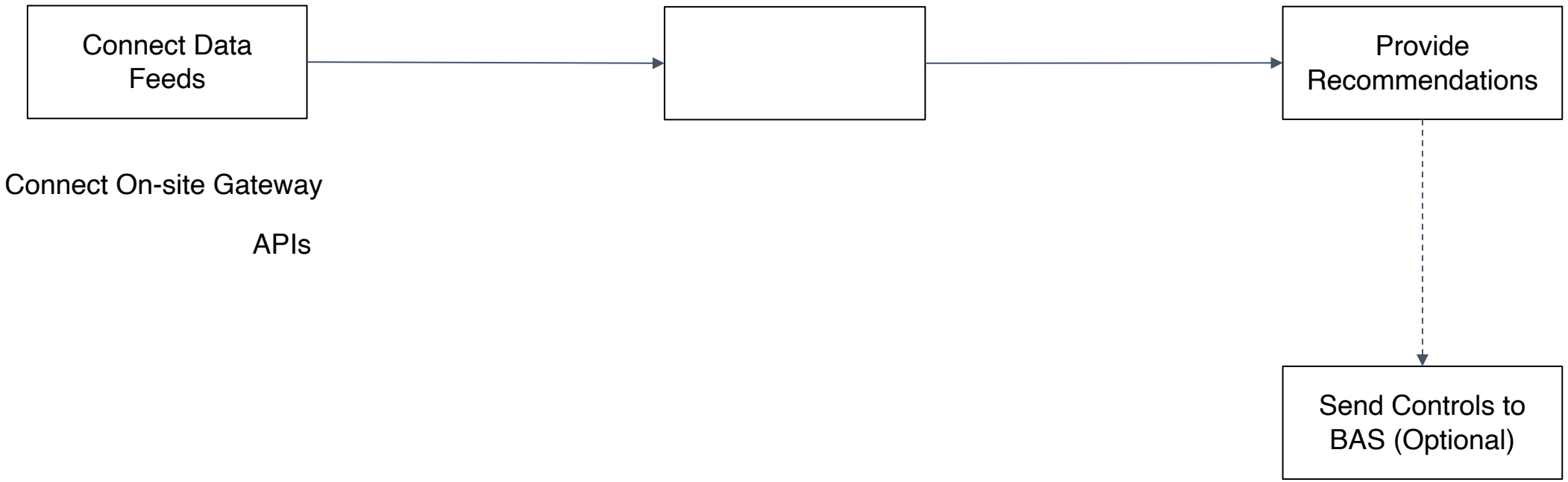
Ambient aims to be a leader in data science solutions for CRE optimization



Best in class in service, cost, and customization

	Ambient Systems Data Science Subscription	Data Science Professional Services	Data Science In-House Teams
Service	On-demand Subscription	Limited Time Project	Hard to recruit Requires on-boarding Training needed Management overhead High turnover risk
Cost	Affordable	Expensive	Expensive
Customization	Specialized	Generic	Requires multiple experts to tune Technical uncertainty and risk Expensive and complex IT vendor management process Cybersecurity risks

Part 1 - Platform: end-to-end CRE property optimization and decarbonization

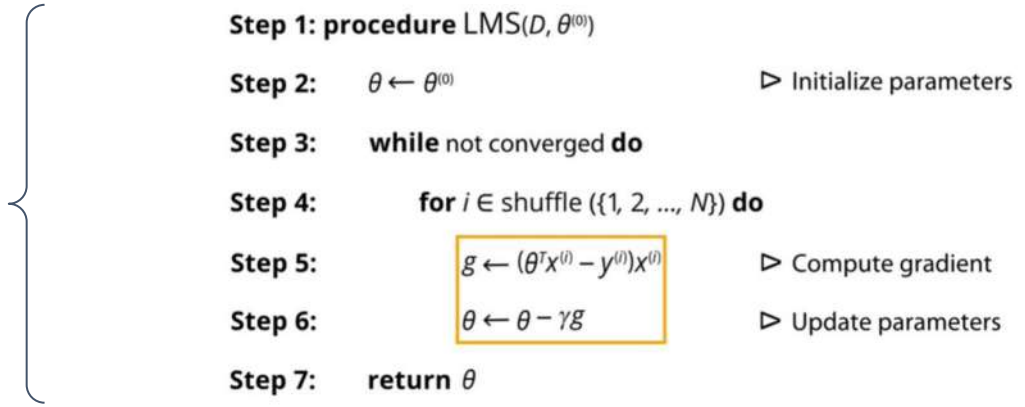


Part 2 - Models: latest models tailored for your occupiers

Key Steps:

1. Identify variable correlations (e.g., Energy Consumption and Temperature)
2. Formulate objective function (e.g., minimize energy consumption)
3. Formulate constraints functions (e.g., IAQ, CO2, Relative Humidity, Thermal Comfort)
4. Optional: Assign objective vector weights
5. Perform multi-objective optimization leveraging (i.e., vector optimization)

Stochastic Gradient Descent



Tailored CRE data and insights platform, balancing productivity with sustainability

Insights



Portfolio

- Portfolio Performance
- Leasing
- Transactions



Facilities & Operations

- Decarbonization
- Operational Performance
- Maintenance & Repairs
- Energy Efficiency



Space & Workplace

- Workplace Performance
- Occupancy
- Utilization

Platform

CRE Data Model	API Connections	Data Engineering	Data Visualization	Notifications

Applications

Use cases and Outcomes

Ambient supports a broad a range of CRE assets

			
Industrial	**Retail**	**Offices**	**Transportation & Logistics**
Manufacturing R&D Life Sciences and BioTech	Malls Shopping Centers Grocery Stores	High Rise Low / Mid Rise Data Centers	Public Transit Airports Warehouses and DCs

Outcomes include improvements in both operational efficiency and competitive advantage

Enhance Operational Efficiency	Enhance Competitive Position
• Reduce FM, technician, and skilled trades costs • Improve Energy Efficiency • Enhance Asset Value • Reduce (Operational & Embodied) • Improve occupancy rates • Enhance asset yields and profitability	• Portfolio investment performance • Quality of service provided to occupiers • Investor liquidity and willingness to invest • Fund terms (including fees) • Brand recognition • Business reputation



15% to 25%

Energy Savings



20% to 40%

Reduction in



10% to 20%

Increase in Equipment Uptime



3X

Manhour Time Savings

Empowering Leading Real Estate Asset Managers and Operators in Asia and Globally

Firm	HQ	AUM in Asia
GLP	Singapore	$65B
IGIS Asset Management	Seoul, Korea	$36B
ESR	Hong Kong	$26B
Blackstone	New York, USA	$26B
Brookfield Asset Management	Toronto, Canada	$26B

Case Study

AgTech POC in Singapore


Indoor Farming requires coordination of complex parameters, such as lighting, moisture, and CO2, requiring constant monitoring and adjustments to ensure HIGH-QUALITY output

Too Many Manual Systems = Labor Intensive



Savings Potential Through Increased Human Productivity

Utilities
$3



Rent
$30



Human Capital
$300



Only 51% of Indoor Farms Are Profitable (Agrilyst, 2019)

Undesired Interactions Between Building Conditions and Crops

(External Building Conditions)

Temperature

(Vertical Racks Ecosystem)

Soil Moisture

Lighting

Nutrients



Humidity

Building Conditions affect crops

- Inconsistent Environment
- Habitat for Mold and Pest
- Affect level of photosynthesis

Undesired interactions between buildings and crops



Strawberries growing in varying levels of
Temperature
Humidity
CO2 Levels





Habitat for pests

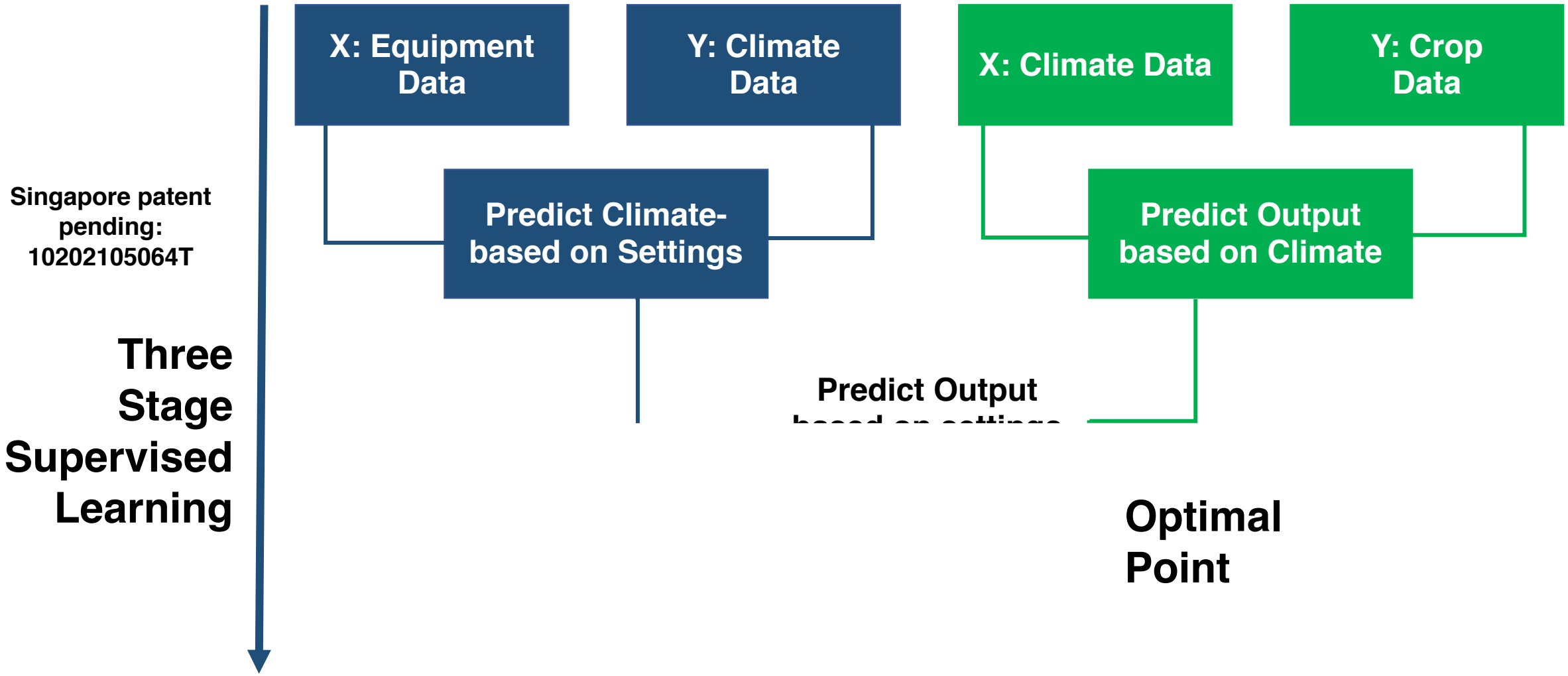
Insights from POC Pilot

Parameter	Finding	Potential Improvements
	CO2 at 25% of the optimal	• Increased photosynthesis by up to 75% • Earlier flowering • Higher fruit yields by up to 30%
	Temp at 70% of the optimal	• Increased leaf and petiole growth • Greater sucrose • More biomass at the roots

Ambient proprietary method provides equipment setpoints to achieve optimal targets

Time	Thermostat 1	Thermostat 2	Fan 1	Fan 2
May 28 – 2 pm	14 C	12 C	2 ACH	5 ACH
…				

Patent: Computer System for Management of Buildings with Indoor Farms



Trade Secret: Operational Data on Creating an Optimal Indoor Climate

Climate Data

	Parameter
Current POC	Air Temperature
	Relative Humidity
	CO2
	Spectrum Light

Crop Data

	Parameter
Full-scale Plot	Soil Temperature
	Soil Moisture
	pH level
	Electrical Conductivity (EC_w)
	Crop specific parameters

Building Data

Equipment Settings
• Equipment setpoints • Light schedules • Chiller Plant settings • AHU Fan settings • Auxiliary Devices

Current Solutions are Lacking

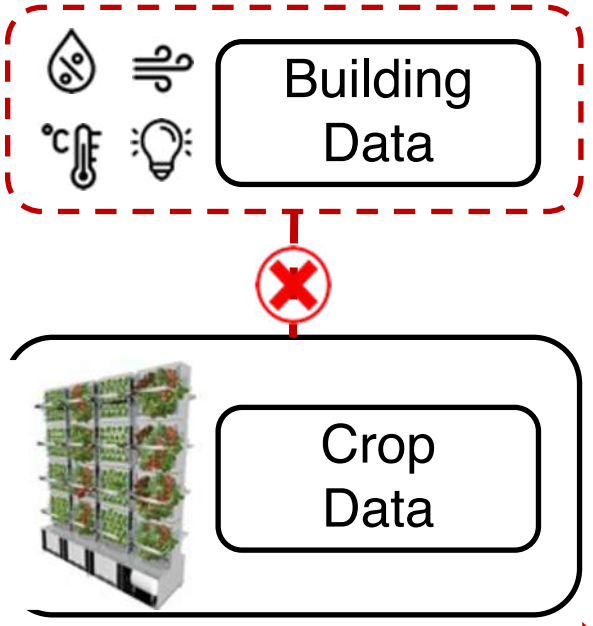
Farm systems do not collect building data



Container farms not built for scale



Building solutions are too general



Solution: AI Platform for AgTech Facilities





Sense



Optimize



Control



Singapore patent pending: 10202105064T

1. Connected

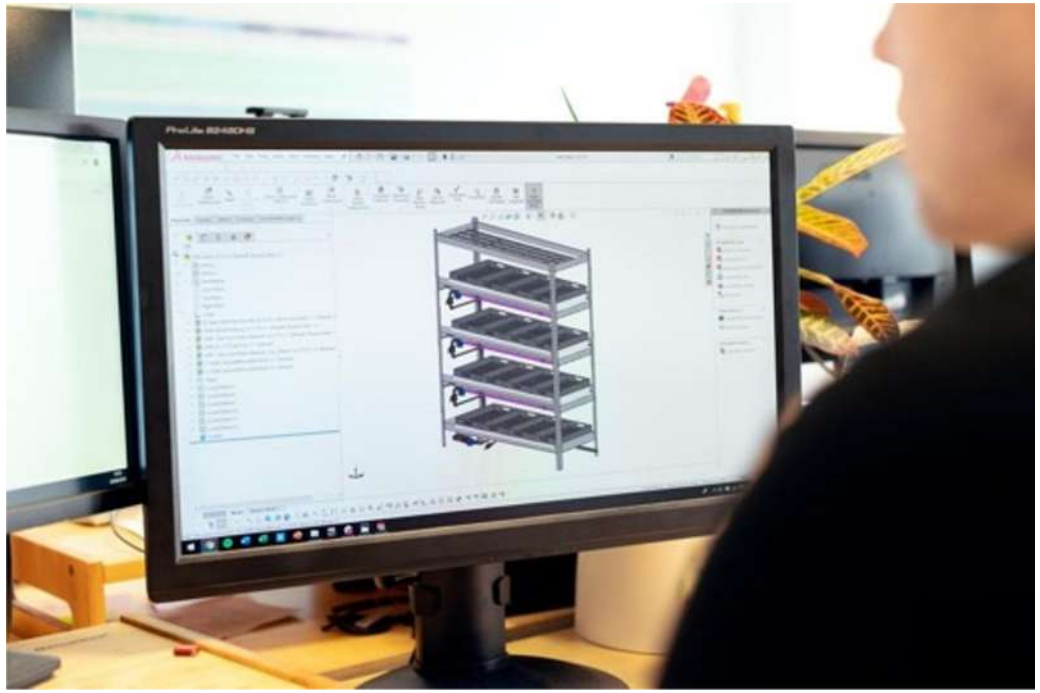
Integrated with essential equipment and control systems

Seamless Building Integration

Connect the indoor farming system to the existing building management system – or have Ambient implement them based on your requirements. Enjoy full connectivity across the farm and perform real-time environmental adjustments based on crop.

2. Centralized

Important data in one easy-to-access place



Monitoring In Real-Time

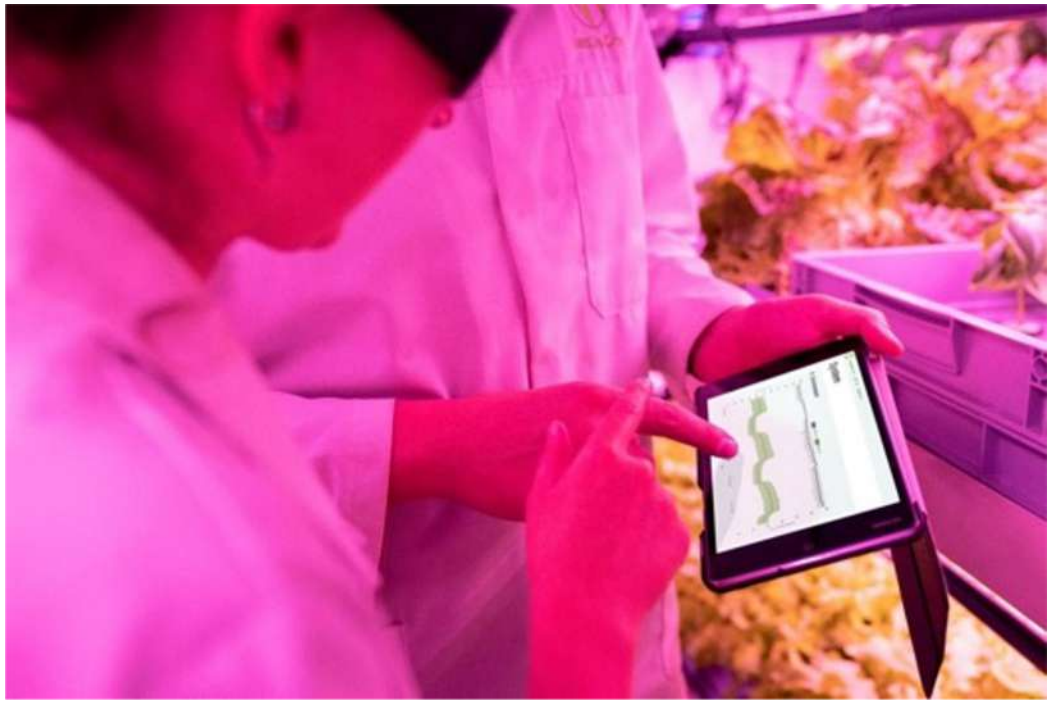
Access real-time data through online dashboards and securely archive all critical operational information. Start measuring most important crop parameters like temperature, relative humidity, and CO2.

3. Automated

Leverage combined power of AI and Human Experts

Grow With Data Insights

Leverage latest models to improve the harvest and reduce operational costs. Access curated insights by our network of crop-scientists who can help you tun—in your farm operations, enabling better crop health and more accurate harvest timing predictions.



Data-Driven Automation

Climate Data



Crop Data



Equipment Settings



	Parameter
Monitor	Air Temperature
	Relative Humidity
	CO2
	VOCs

	Parameter
Crop Insights	Soil Temperature
	Soil Moisture
	pH level
	Electrical conductivity (EC_w)
	Crop specific parameters

Sample Action
• Adjust temperature • Adjust ventilation • Adjust lights • Turn on fogging machines • Turn on CO2 regulator • Balance irrigation and humidity levels



+24%

energy savings,
man-hour savings,
enhanced crop performance

Break-even in

6 to 12 months

Usage-based Pricing Plan

One-time Implementation Cost
Installation of sensors, networks, and equipment integration

by Project Basis

Basic
SGD $500 per week

- Custom data analytics

2-3 Data Feeds

Advanced
SGD $1,000 per week

- Custom data analytics
- Customized AI module

4-5 Data Feeds

Premium
SGD $2,000 per week

- Custom data analytics
- Industry vetted AI module
- Cybersecurity enhancements

6-7 Data Feeds

Traction



Working with three AgTech Companies in Singapore

Approved by industry leaders in building technology

POC Project Proposal

Ambient Systems & CapitaLand

Simple steps towards effective POC execution and results

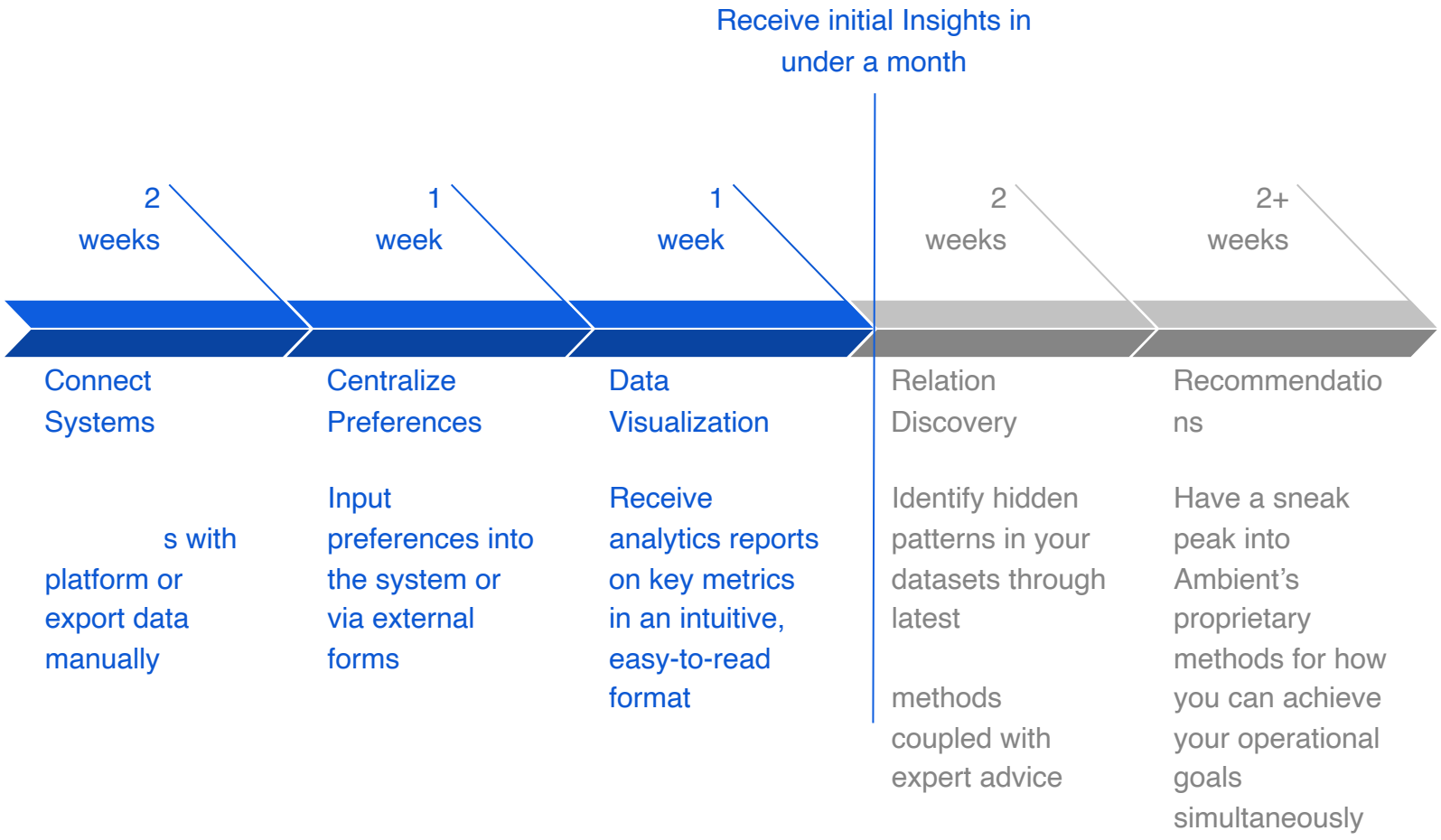
Arrange a Consultation

Identify priority ML use-cases

Establish POC objectives, timeline, and resources

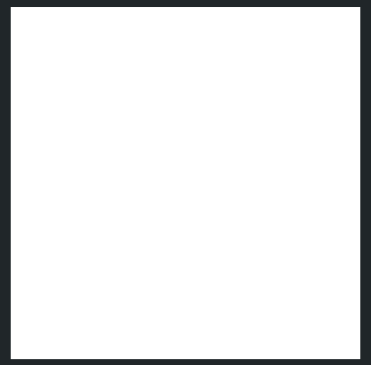
Deploy a small-scale pilot in production (e.g., a building section)

POC duration can range 3 to 4 months with initial insights after 2 weeks from setting up the system

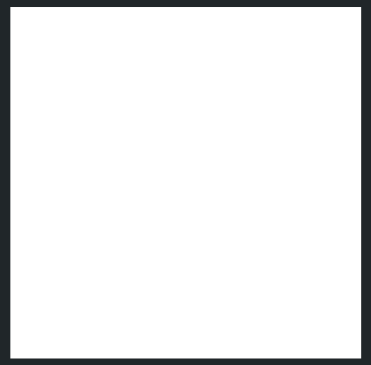


ABOUT US

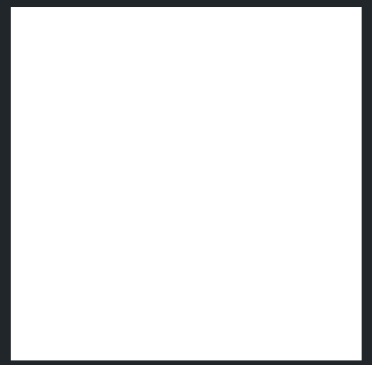
Our Team



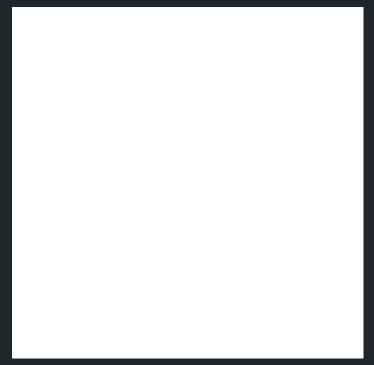
CEO
Deep tech entrepreneur with 3+ years experience in [redacted] and clean tech



UX/UI Designer
Service & UIUX Designer with background in VC & Industrial Design

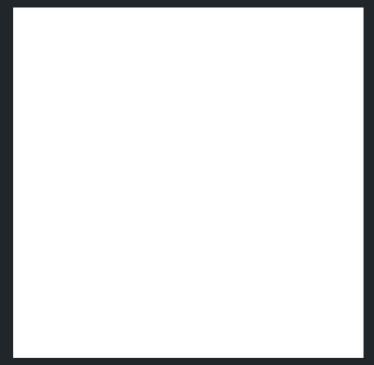


Data Scientist
[redacted] engineer with experience building cross-industry AI applications at Meta and AWS

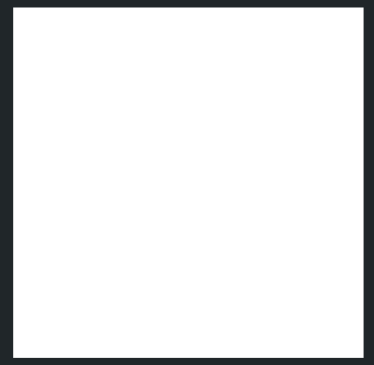


[redacted] Developer
Experienced developer of gaming and advanced web-based applications

Mentors & Advisors



Dr. [redacted]
Experienced technology executive, investor, and educator with 35 years of leadership experience at Oracle and HP

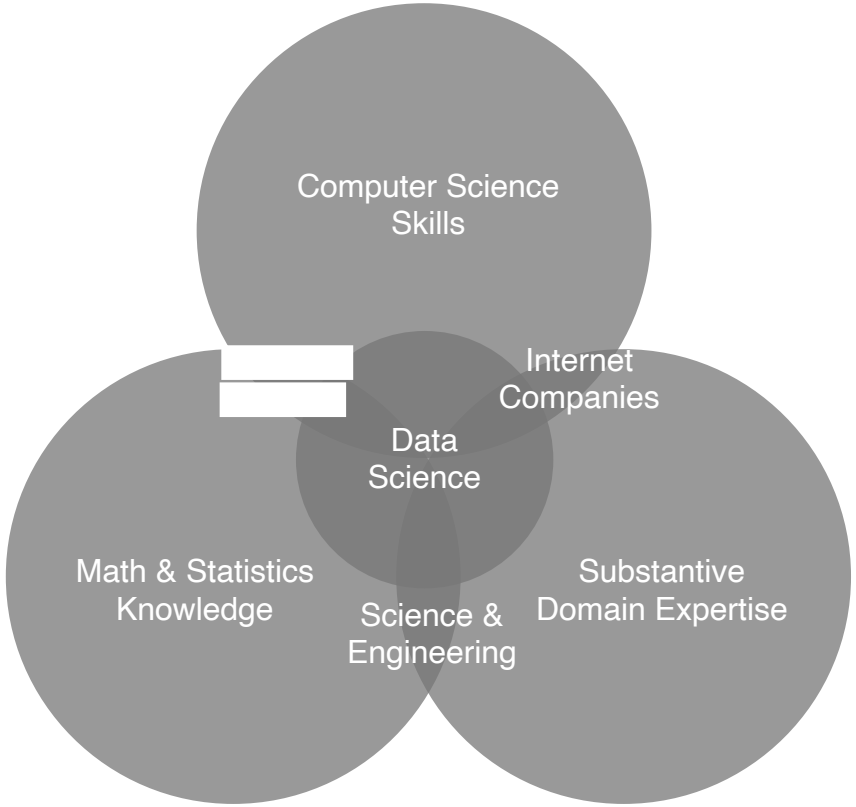


Dr. [redacted]
Experienced private equity investor and University professor in global innovation and entrepreneurship

Extra Slides

FAQ

Q: Why are Data Scientists hard to find?



Niche Requirements
Shortage of data scientists who can optimize multiple intelligent systems

Competitive Landscape
Only around 2k data scientists in Singapore and around 50k in US, with majority working for tech giants

Data Science Teams can require over 10 roles, making them too costly for most organizations

Q: What do Data Scientists do?

General Activities

- Find internal and external data to identify new opportunities or risk areas
- Develop an understanding of user needs
- Communicate findings to stakeholders
- Implement data-driven insights
- Keep a pulse on the latest technical and industry developments and how they can enable smarter business processes



Specific to Real Estate

1. Control energy efficiency
2. Balance energy savings with environmental considerations (healthy environment (IAQ), CO2, etc.)
3. Create an environment for driving productivity and revenues
4. Streamline compliance (regulations, certifications, and standards)

Q: What are Cyber Physical Systems?

Cyber Physical Systems (CPS) Components

Internet

+

Electronic Devices (IoT)

+

Physical Surroundings & Equipment

+

Users

Cyber Physical Systems (CPS) Applications

Commercial Buildings

Industrial Buildings

Transport Infrastructure

Healthcare Facilities

Smart Homes

Utilities (water, electricity, natural gas)

Q: What are Ambient's Data Science Capabilities?

Data Engineering **(Input)**	**(Use Cases)**	Data Visualization **(Output)**
• Data Selection • Data Preprocessing • Data E hment • Feature Engineering • Data Transformation • Exploratory Data Analysis	• Predictive Maintenance • Demand Forecasting • Image and Video Analytics • Inventory Optimization • Recommender Systems • **Graph Mining (Discovering insights in real world networks)*** • **Equipment Calibration*** • **Business Objective Balance and Optimization***	• Moving average • Box plots • Histograms • Contours and Density Contours • Hexbin • Q-Q Plot • Parallel Sets • Treemap and hierarchy graphs • Maps and map tiles

***Only offered by Ambient**

Technologies we Use



Q: Do you offer hardware, sensors, or BMS?

We don’t sell them directly but we can help you evaluate vendors to make sure they meet your business needs and you can access the data you need.

Q: Will you recommend AC System upgrades?

Yes, we will provide recommendations where we see room for improvement.

As with the BMS, we do not sell it ourselves.

Q: How long to see the results?

The accuracy and insights will improve over time.

But you can expect initial insights within two weeks after connecting the relevant data feeds.

Q: Is my data secure?

Yes.

We work with leading vendors, such as AWS and C3.AI, to power our platform and models

We also follow national and international regulations on data protection and data privacy.